Chapter 9 • Similar Figures

Solved Exercise 9.2

1. Find the ratio of the areas of similar figures for each given ratio of corresponding lengths.

(i) Lengths 1:3
⇒ areas $1^2:3^2=1:9$.
(ii) Lengths 3:4
⇒ areas $3^2:4^2=9:16$.
(iii) Lengths 2:7
⇒ areas $2^2:7^2=4:49$.
(iv) Lengths 8:9
⇒ areas $8^2:9^2=64:81$.
(v) Lengths 6:5
⇒ areas $6^2:5^2=36:25$.

2. Find each unknown from the five similar-figure diagrams.

(i) $A_2 = 240(6/10)^2$
$= 86.4\ cm^2$.
(ii) $A_2 = 60(20/15)^2$
$= 106.67\ cm^2$.
(iii) $A_1 = 18(3.6/5.76)^2$
$= 7.03125\ cm^2$.
(iv) $A_1 = 96(15/12)^2$
$= 150\ cm^2$.
(v) Small area=$3\frac{4}{7}=\frac{25}{7}\ cm^2$. Area ratio $(\frac{25}{7}):63=25:441$, so length ratio=5:21. Thus $\frac{3}{x}=\frac{5}{21}$
x=12.6 cm.

3. Area ΔABC=36 cm², AB=6 cm and BD=4 cm, with BC∥DE. Find areas of ΔADE and trapezium BCED.

AD=AB+BD=10 cm. Since ΔABC~ΔADE, area ratio=$(AB/AD)^2=\frac{6}{10}^2=\frac{9}{25}$

Area ADE = $36\times\frac{25}{9}$
$= 100\ cm^2$; trapezium area
$= 100-36$.
(a) $100\ cm^2$
(b) $64\ cm^2$.

4. ΔABC and ΔDEF are similar with scale factor k=3 (first to second); area ABC=50 cm². Find area DEF.

Area ABC/Area DEF = 3^2
$= 9$.
Area DEF=$\frac{50}{9}=5\frac{5}{9}\approx5.56\ cm^2$.

5. Similar quadrilaterals ABCD and EFGH have k=$\frac{1}{4}$ (first to second); area ABCD=64 cm². Find area EFGH.

$\frac{64}{\text{Area}}$ EFGH = $(\frac{1}{4})^2$
$= \frac{1}{16}$
Area EFGH=64×16=1024 cm².

6. Areas of two similar triangles are 16 and 25 cm². Find corresponding-side ratio.

$\sqrt{16}:\sqrt{25}=4:5$.

7. Areas are 144 and 81 cm²; the larger base is 30 cm. Find the smaller base.

Length ratio small:large=9:12=3:4, so smaller base=$30\times\frac{3}{4}=22.5$ cm.

8. A larger regular heptagon has sides 1.7 times the smaller; smaller area=100 cm².

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Larger area = $100(1.7)^2$
$= 289\ cm^2$.

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